

Department of English
End Semester Examination, 2025
(For the 2nd Semester Students of all the Departments of the Engineering Faculty)
ENGUGHU01: Communicative English

Full Marks: 80

Time: 3 Hours

Section I

1. Answer ANY TEN of the following questions:

2x10=20 Marks

1. What are the channels of communication?
2. What is the basic difference between general communication and professional communication?
3. What is intra personal communication?
4. Exemplify a barrier in effective listening.
5. Give an example of Grapevine communication.
6. What is code?
7. How many diphthongs are there in English language?
8. What is 'LSRV'?
9. What is phoneme?
10. What is *lingua franca*?
11. What is semantic gap?
12. What is skimming in reading?
13. What is homophone? Give an example.
14. What is phrase? Give an example.
15. What is monologue?
16. Give two examples of idioms in English language.

Section II

2. Answer ANY FOUR of the following questions: (word limit 200)

5x4=20 Marks

1. What are the characteristics of a good communication?
2. What is business communication? What are the methods of business communication?

Ucan
Course
Comment
Feedback.

work pr
reduces

discuss the barriers of effective listening

Write a short dialogue between two friends discussing the price hike.

What is the difference between summary and paraphrase?

Write a short report on the metro rail in Kolkata.

What are the important features of close reading?

Section III

3. Answer ANY FOUR of the following questions. (Word limit 400)

10X4=40 Marks

1. Define Communication. What are the functions of Communication? (5+5)

2. Define and discuss the differences between Verbal Communication and Non-verbal Communication. (10)

3. Define Grapevine Communication. What are the demerits of Grapevine Communication? (5+5)

4. Explain the different flows of communication in the business organisations. (10)

5. Attaching a Resume, write a job application for the post of a librarian in a university. (10)

6. Mention the different Presentation strategies of Speaking Skills. (10)

7. Write a report of the Annual Teachers Day celebration in your university. (10)

Psychological
Cultural
Perceptual
Language
Physical
Regulator
Motivator
Informative
③
②

topics

Full Marks: 80
Year: I

Instructions:

1. Mention the question number clearly and write all the parts of a question at one place.
2. Draw circuit/figure/phasor diagram wherever applicable (including numerical).
3. Make suitable assumptions wherever necessary. Symbols and notations have their usual meanings.

1. In an R-L circuit, $L = 2 \text{ H}$ and $R = 4 \Omega$. Find the current at $t = 1 \text{ s}$ after switch is closed to a 10 V DC source.
2. What is the role of brushes in the DC machine?
3. Give two basic speed control schemes of DC shunt motor?
4. Differentiate between capacitive and inductive reactance.
5. State the difference between linear and non-linear networks.
6. Define power factor. What is its significance?
7. Can DC voltage be applied to transformers?
8. A transformer has a primary voltage of 220 V and secondary voltage of 110 V . If the primary turns are 1000 , find the number of secondary turns.
9. A 230 V DC motor draws 10 A and its back EMF is 210 V . Find the armature resistance.

Answer any five questions

5 x 12 = 60 Marks

2. a) Determine the three mesh currents as shown figure in Fig. 1.
- b) Use nodal analysis for computing current through the 30Ω resistor in Fig. 2

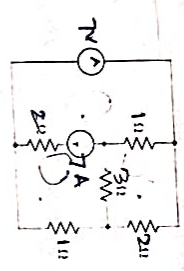


Fig. 1.

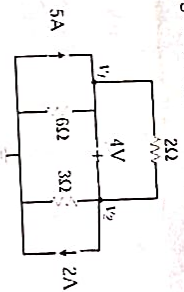


Fig. 2.

- a) Find the Thevenin equivalent circuit across terminals 'a' and 'b' as shown in Fig. 3.
- b) Find the Norton equivalent circuit across the terminals a-b of the circuit shown in Fig. 4.

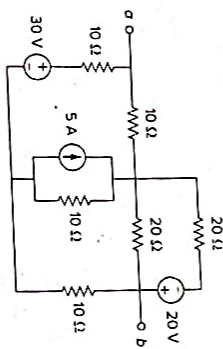


Fig. 3

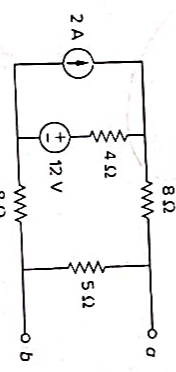


Fig. 4

6/16

60, 48, 48

In R-L circuit with DC excitation (driven circuit) determine inductor voltage and inductor current. 6+6
In an R-L circuit with the time constant of 1.25s, inductor current increases from the initial value of zero to the final value of 1.2A. i) Calculate the inductor current at time 0.2s, 0.4s and 1.2s. ii) Find the time at which the inductor current reaches 0.3A, 0.6A and 0.9A

How does the back emf regulates the armature current in a D.C. Motor? Explain with relevant equations. 6+6
A 220 V series motor takes 10A and runs at 600 rpm. The total resistance is 0.8 Ω . At what speed will it run, when a 5 Ω resistance is connected in series, the motor taking the same current at the same supply voltage? 6+6

Derive an expression for the electromotive force induced in the secondary winding of a transformer. 6+6
A 17.5 kVA, 460/115 V single phase (50 Hz) transformer has primary and secondary resistance of 0.38 Ω and 0.02 Ω , respectively and the leakage reactance of these windings are 0.82 Ω and 0.06 Ω , respectively. Determine the voltage to be applied to the primary side to obtain full-load current with the secondary winding short circuited. Neglect the magnetizing current. 10

A circuit consisting of resistance of 20 Ω , an inductance of 0.05 mH connected in series. A supply of 230 V at 50 Hz is given to the circuit. Find the impedance, current, power factor and power consumed in the circuit. 6+6
Justify, why pure inductor does not consume any power when connected across single phase A.C. supply?

Draw the phasor diagram for lagging p.f. load in an ideal transformer. 4+8
a) A 10 kVA, 2300V/230 V, 50 Hz single-phase transformer has the following test data:
Open-circuit test (L.V. side): 230 V, 2 A, 100 W
Short-circuit test (H.V. side): 100 V, 4.35 A, 150 W

Calculate:
i) The equivalent circuit parameters referred to L.V. side
ii) The efficiency at full load, 0.8 power factor lagging
iii) Voltage regulation at full load, 0.8 power factor lagging

$$E_p = \frac{\phi Z N P}{60 A}$$

End Semester Examination - 2025 (Spring)

MATUGBS02

(Engineering Mathematics II)

Semester - II (4 Year B.Tech)

Full Marks - 80

Time - 3 Hours

Notations and symbols have their usual meaning.
Use separate answer script for Group-A and Group-B

GROUP-A

Section A. Answer ALL questions. Each question carries TWO mark.

1. ~~(a)~~ Define skew symmetric matrix with example.

~~(b)~~ Show that any square matrix can be written as sum of two matrices, one is symmetric and another is skew symmetric.

~~(c)~~ Obtain the determinant of the matrix

$$A = \begin{bmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{bmatrix}$$

~~(d)~~ Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, B = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$$

show that $AB = BA$.

~~(e)~~ Define rank of a matrix.

Section B. Answer any THREE questions. Each question carries 10 marks.

2. (a) Define echelon form of a matrix with example. Reduce the echelon form of matrix A, where

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 4 & -1 & 5 \\ 2 & 0 & 6 \end{bmatrix}$$

3+6+1

Hence comment on the rank of A

3. (a) Show that $\text{Det}(A) = (a-b)(a-c)(b-c)$, where

$$A = \begin{bmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{bmatrix}$$

$$\begin{bmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1} \begin{bmatrix} a^2 & a & 1 \\ b^2 - a^2 & b - a & 0 \\ c^2 - a^2 & c - a & 0 \end{bmatrix}$$

4

4x4

(b) Find two matrices P and Q such that $PQ = I_3$, where

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 3 & 0 \\ 6 & 2 & 3 \end{bmatrix}$$

6

(3) Let $D_n = \text{Det}((u_{ij}))_{n,n}$, where $u_{ij} = a^{i-j}$, $a > 0$, show that $D_n = (1 - a^2)^{n-1}$.

4

(4) Solve the following system of equations using Cramer's rule:

$$\begin{aligned} x - 2y + 2z - 2 &= 0 \\ 2x - y - 2z - 1 &= 0 \\ 2x + 2y + z - 7 &= 0. \end{aligned}$$

6

(a) If A is a non-singular matrix then show that (i) A^{-1} (ii) $\text{Adj}(A)$ is also non-singular. Find the inverse of the matrix A , where

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{bmatrix}$$

(2+2)+6

(a) Define orthogonal matrix. Check whether the below matrix is orthogonal or not.

$$A = \begin{bmatrix} 2/3 & 1/3 & 2/3 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/3\sqrt{2} & -4/3\sqrt{2} & 1/3\sqrt{2} \end{bmatrix}$$

$$\begin{aligned} &1(28 - 12) \\ &-1(14 - 12) \\ &+2(6 - 2+8) \\ &16 - 2 - 12 \\ &2 \end{aligned}$$

GROUP - B (40 marks)

Answer all the following questions:

(a) The solution of the given exact differential equation $(\frac{1}{y} - \frac{2}{x})dx - (\frac{x}{y^2} - \frac{2}{y})dy = 0$ is

- i. $\log(\frac{x}{y}) + \frac{x}{y} = c$
- ii. $\log(\frac{x}{y}) + \frac{x}{y} = c$
- iii. $\log(\frac{x}{y}) + \frac{x}{y} = c$
- iv. $\log(\frac{x}{y}) + \frac{x}{y} = c$

(where c is arbitrary constant)

(b) Which of the following options is integrating factor of the given differential equation $y(2xy + e^x)dx - e^x dy = 0$

$$y(2xy + e^x)dx - e^x dy = 0$$

1. $\frac{1}{x}$
 ii. $\frac{1}{x^2}$
 iii. $\frac{1}{x^3}$
 iv. $\frac{1}{x^4}$

Which of the following options is the particular integral of the given differential equation $(D^2 - 1)y = 2e^x$ where $D \equiv \frac{d}{dx}$

- i. xe^x
 ii. x
 iii. e^x
 iv. e^{x^2}

$\frac{1}{f(b)} \times 2e^x$
 $\frac{1}{x^2 - 1} \times 2e^x$
 $\frac{2e^x}{x^2 - 1}$

Find mod z and amp z (principal amplitude) where $z = \frac{1 + \sqrt{3}i}{1 + i}$

- i. $\sqrt{2}, \frac{\pi}{4}$
 ii. $\sqrt{2}, \frac{\pi}{3}$
 iii. $\sqrt{3}, \frac{\pi}{3}$
 iv. $\sqrt{3}, -\frac{\pi}{4}$

$\frac{2 \cos \frac{\pi}{6} - \sin \frac{\pi}{6}}{2 - 1}$

If $z_1 = 1 + i$ and $z_2 = 1 + \sqrt{3}i$, then the principle amplitude of $z_1 z_2$ is

- i. $\frac{7\pi}{12}$
 ii. $\frac{5\pi}{12}$
 iii. $\frac{3\pi}{4}$
 iv. $\frac{2\pi}{3}$

$(1 + i)(\sqrt{3}i + 1)$
 $\sqrt{3}i + 1 + \sqrt{3} + i$
 $(\sqrt{3} + 1)i + 1 + \sqrt{3}$

Answer any SIX of the following questions:

- (i) Solve the linear differential equation $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$
 (ii) Test for exactness and solve $x^2(1 + y)D^2y + 2x(2 + 3y)Dy + 2(1 + 3y)y = 0$
 (iii) Solve $(x^3 D^3 - x^2 D^2 + 2x D - 2)y = x^3 + 3x$
 (iv) Solve the Bernoulli's differential equation $\frac{dy}{dx} - 3xy^2 = -e^x \cdot 2x^3 y^3$

$\frac{1 + \sqrt{3}}{2} + \frac{2\sqrt{3}}{2}i$
 $\frac{1 + \sqrt{3}}{2} + \sqrt{3}i$

$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{e^x}{x^2}$

Solve: $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = 3x + 1$
 Solve: $\frac{d^2y}{dx^2} - 4y = xe^{2x}$
 Solve: $\frac{d^2y}{dx^2} + a^2y = \sin ax$

Show that the solution of the differential equation (Clairaut's form) $y = px + f(p)$ is $y = cx + f(c)$ where c is an arbitrary constant and hence solve $y = px + p^2 + 2p + 1$, find other two roots of the equation $x^3 - 10x^2 + ax - b$ (where $a, b \in \mathbb{R}$) is $4 + i$, then find (i) If one of the root of the equation $x^3 - 10x^2 + ax - b$ is $4 + i$, then find other two roots and also find the value of a and b .
 OR, (ii) If x, y, z are positive real numbers and $x + y + z = 1$, prove that

$(1 + \sqrt{3}i)(1 - i)$

$8xyz \leq (1 - x)(1 - y)(1 - z) \leq \frac{8}{27}$

$\frac{1 + \sqrt{3}}{2} + \frac{2\sqrt{3}}{2}i$

$1 + i$
 $1 - i + \sqrt{3}i + \sqrt{3}$
 $(1 + \sqrt{3}) + (\sqrt{3} - 1)i$

Use separate answer scripts for each group

Group A

Explain the following terms (answer any FOUR): (a) Path function (b) Irreversible process (c) Molar heat capacity (d) Enthalpy (e) Extensive property (f) Joule Thomson phenomenon

answer any FOUR from the following questions

02. (a) Derive the expression for the work done during the isothermal, reversible expansion of an ideal gas. (b) Draw the pressure-volume (P-V) diagrams depicting the isothermal expansion of an ideal gas, clearly differentiating between reversible and irreversible processes. 2.5+2.5

03. (a) State the thermodynamic condition for equilibrium. (b) Show Joule Thomson effect is an isenthalpic process. 2.5+2.5

(c) Prove mathematically that work done in a reversible process is more than that of an irreversible process. (d) Differentiate between reversible and irreversible processes. 2.5+2.5

(e) Explain the concepts of C_p and C_v for gases. 2.5+2.5

(f) Why is C_p always greater than C_v ? Establish the relationship between them. 2.5+2.5

(g) Find the work done when one mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 25°C. (h) 1 gm of water is converted into steam at the same temperature. The volume of water becomes 1671 ml on boiling. Calculate the change in the internal energy of the system if the heat of vapourization is 540 cal/gm. 2.5+2.5

Group B (Answer any TEN) 5 × 10 = 50

07. Give brief account of i) Polythene ii) Compounding of rubber

08. (a) Why are silicones called inorganic polymers? (b) Discuss the synthesis procedure of linear chain silicones.

(c) What is meant by green chemistry?

(d) Write informative notes on the mechanism of radical polymerisation

09. Write notes on: (a) Polystyrene (PS) (b) Polyvinyl Chloride (PVC).

10. What is meant by the vulcanisation of rubber? How vulcanisation takes place? Write its uses?

11. What is Biopolymers? How is it prepared? Give some importance of Bio-polymers.

12. Write down the importance of green chemistry. Give the preparation of Aspirin from salicylic acid using microwave technique. Why this technique is preferred?

13. Explain with examples: i) Grafting ii) Conducting polyaniline

14. What is Ziegler Natta catalyst with some application. b) How will you prepare Styrene-Butadiene Rubber (SBR). Mention its uses.

15. (a) What are drawbacks of natural rubber? (b) How will you prepare Novolac resin from phenol and formaldehyde with mechanism

16. (a) What is meant by degree of polymerisation? (b) Distinguish between addition and condensation polymerization.

17. How the following are prepared? Mention their uses (a) Teflon (b) LDPE

18. a) How is the Crepe rubber obtained from latex? b) What is meant by degree of polymerisation?

19. a) Distinguish between addition and condensation polymerisation with examples. b) Give a short note on microwave radiation in green synthesis.

$$\textcircled{1} \left(\frac{f-1}{2} \right) \ln \frac{1}{1-x}$$

(1N) (1N)

Aliah University

End-Semester Examination-2025

BTech 2nd Semester [Civil Engineering and Mechanical Engineering]

Paper Name: Programming for Problem Solving

Paper Code: CSEUES01

Full Marks: 80
Time: 3 Hrs

Group - A

Fill in the blanks with correct alternative-

- C compiler is _____ [case sensitive/ not case sensitive].
- A header file keeps _____ [declaration/ definition] of library functions.
- 'xy' is a valid variable name in C compiler. _____ [true/ false]
- What is the answer of: $59 \div 3$? _____ [-141]
- scanf function can read _____ [only single word/ multi word] strings.
- The sum of two hexadecimal numbers 87 and 78 is = _____ [165/ 0F]
- sizeof() is a/an _____ [operator/ function].
- Continue keyword is for _____ [quitting the loop/ skipping the condition].
- Array name could be used as a pointer. _____ [true/ false]
- Size of 'p' in the declaration: int *p; is same as size of int. _____ [true/ false]

Group - B
(Answer any 5 questions)

- Write a short note on various logical, relational, bitwise operators in C programming. (Understand, CO3) (5)
- Perform the multiplication (all digits are in Octal): 76×53 (Apply, CO2) (5)
- How while() and do-while() loops are different from one another? Explain with example. (Analyze, CO3) (5)
- Write a program in C to find out maximum and minimum number out of three integer values, by using conditional operator. (Apply, CO1, CO3) (5)
- Write a C program to insert some integer values in an Array and then print the average of them. (Apply, CO1, CO5) (5)
- Explain the basic string standard functions with proper examples. (Remember, CO4) (5)
- Write down outputs for the programs below: (give explanation) - (Evaluate, CO5) (5)

```
#include<stdio.h>
#include<conio.h>
int main()
{
    int a[]={4,15,6,88,44,31,22,10};
    int *p=&a[3];
    printf("%d",*(p--));
    printf("%d",*p+3);
    return 0;
```

Handwritten calculations and code snippets:

- Octal multiplication: $76_8 \times 53_8 = 4080_{10}$
- Array program output: 43
- String function examples: $2610, 300, 310, 44$
- Mathematical calculations: $812, 49 \times 3, 7 + 42, 2 \times 3 \times 3$
- Hexadecimal calculations: $87_{16} + 78_{16} = 0F_{16}$

$$(1-n) + 100 \times u$$

$$3^3 + 8^2 + 8^1 + 8^0$$

$$27 + 64 + 8 + 1 = 100$$

Group - C
(Answer any 4 questions)

$$\frac{64}{2} = 32$$

$$\frac{32}{2} = 16$$

$$\frac{16}{2} = 8$$

$$\frac{8}{2} = 4$$

$$\frac{4}{2} = 2$$

$$\frac{2}{2} = 1$$

9) Define Recursion. Write a program in C to find out factorial of any integer number by using Tail-recursion.
(Apply, CO4)

$$5 \times 8^3 + 1 \times 8^2 + 5 \times 8 + 2$$

$$5 \times 512 + 64 + 40 + 2 = 2638$$

10) Write a C program to check whether a number is Perfect or not.
(Apply, CO3, CO4)

11) Define function. What are the advantages of using function? Write a program in C to check whether an integer number is prime or not. Use separate function for the same.
(Analyze, CO1, CO6)

26	38	4	45	67	17	9
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12) Write a note on - Switch and case. What is the role of 'break' in switch-case? Explain with example.
(Understand, CO3, CO4)

13) Convert the decimal number: -39 into its correspondent Sign-Magnitude, 1's complement and 2's complement forms.
(Apply, CO2) [2+2+2+2=10]

14) Define Pointer. Explain the concepts of 'pointer to array' and 'Array of pointers' with suitable examples.
(Analyze, CO5)

Write a program in C to reverse a string without using library functions.
(Apply, CO2, CO4) [2+4+4=10]